

Lecture Notes

EC226 Term 1 2025/26

Multiple variable regression model

Readings:

- Stock and Watson (2003): Chapters 6 + 7
- Dougherty (2016): Chapter 3
- Wooldridge (2013): Chapters 3 + 4 + 5

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1 Introduction

What are the consequences of simply looking at a bivariate relationship? The University of California, Berkeley was sued for bias against women who had applied for admission to graduate schools. The admission figures for the fall of 1973 showed that men applying were more likely than women to be admitted, and the difference was so large that it was unlikely to be due to chance.

	Applicants	% admitted
Men	8442	44%
Women	4321	35%

A regression of % admitted on the variable indicating whether the applicant was female gives:

$$\text{Admit} = 44.0 - 9.00 \text{ Women} \\ (0.910)$$

The coefficient on *Women* shows this group are 9% points less likely to be admitted and this coefficient is significant:

$H_0 : \beta = 0$ vs $H_1 : \beta < 0$ with critical value $z^{0.01} = -2.32$

$$t = \frac{-9.00 - 0}{0.910} = -9.89 \Rightarrow \text{Reject } H_0$$

However, could it be that we are missing an important (additional) variable? Suppose:

Department A receives 150 applications, 100 from men and 50 from women

Department B receives 150 applications, 50 from men and 100 from women.

Department A has an acceptance rate of 80%,

Department B has an acceptance rate of 20%.

Then Department A admits 80 men and 40 women, whereas Department B admits 10 men and 20 women. Overall, 90 men and 60 women are admitted, giving an acceptance rate of 60% for men and 40% for women.

In Berkeley when looking at applications by Department we have:

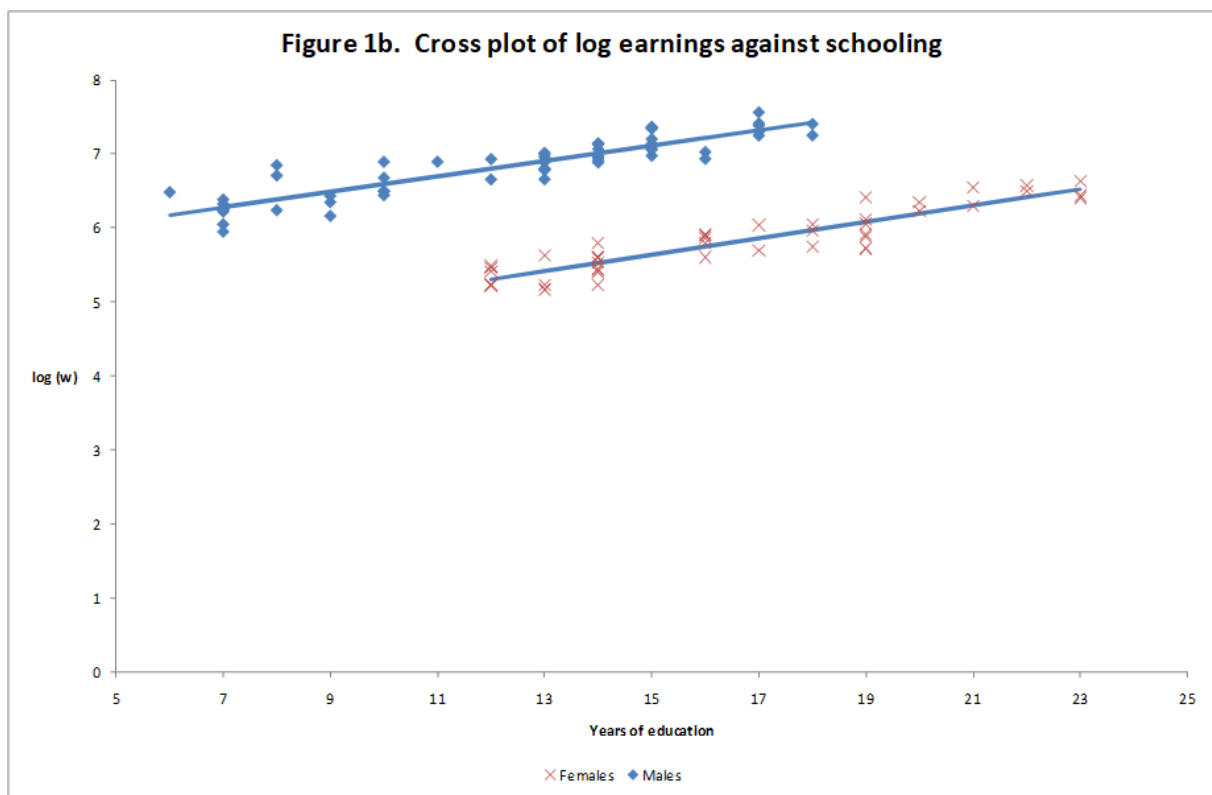
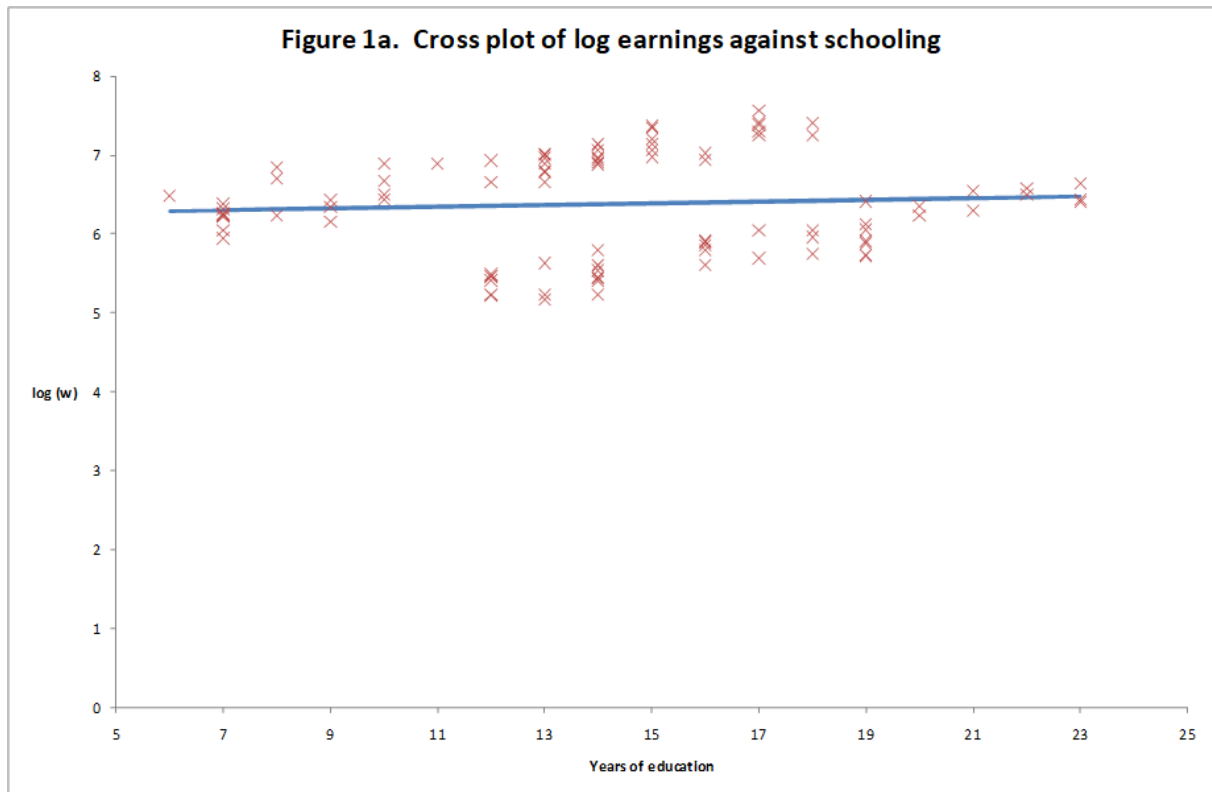
Course	Men		Women	
	Applicants	admitted(%)	Applicants	admitted(%)
A	825	62	108	82
B	560	63	25	68
C	325	37	593	34
D	417	33	375	35
E	191	28	393	24
F	272	6	341	7

So in each Department there is little bias (except perhaps Department A).

However, women tend to apply to more competitive departments (C, E and F) with low admission rates. So in estimated equation for *Admit* above we did not include an important variable, Department of application, and this gives a BIASED and misleading coefficient on women!

Now consider Figure 1a, which plots $\log(w)$ (logged earnings) against Year of education and there is no apparent relationship. Estimating a 2-variable model of $\log(w)$ on Years of education by OLS yields an estimate of a slope coefficient of 0.01, with an estimated standard error on this coefficient of 0.016 (hence the slope coefficient is insignificant).

What is unclear from Figure 1a is there are two types of individuals in the sample (e.g. males and females). In Figure 1b, we re-plot Figure 1a, but identifying males (diamonds) and females (crosses) separately and here there is clearly a strong positive relationship between $\log(w)$ and Years of education for the two groups separately, The problem was that in Figure 1a we did not take look into the effect of gender. What we really want to know is the effect of an additional year of education, for a given (conditional on) gender.



Looking only at the diamonds (males) – we get an estimate of the effect of an increase in Years of education and this is represented by the thick solid line (estimate = 0.10). Looking at the crosses (females) – we can get an estimate of the effect of an increase in Years of education and this is represented by the thin solid line (estimate = 0.11). So why did we get an initial estimate of 0.01? This answer is that, in this case, females have more schooling than males, but females are generally paid less than males – so when we look at a plot of Years of education and $\log(w)$ s the more highly educated females are receiving lower pay (compared to the less educated males). Therefore we need to potentially look at including the influence of all variables that might affect earnings in order to get the “true” effect of education on earnings.

2 Deriving the OLS Estimators

We now extend the analysis in the previous handout to the more realistic case in which we believe the random variable of interest, y , depends upon a number of different factors, $x_{1i}, x_{2i}, \dots, x_{ki}$, and these are exogenous variables. The true but unknown relationship is defined as being

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i, \quad i = 1, 2, \dots, n \quad (1)$$

where we maintain the Classical Linear Regression Model (CLRM) assumptions of the previous handout:

1. $E(\varepsilon_i | x_{1i}, \dots, x_{ki}) = 0$ i.e. $E(\varepsilon_i | \mathbf{x}) = 0$.
2. $Var(\varepsilon_i | \mathbf{x}) = \sigma^2$
3. $cov(\varepsilon_i, \varepsilon_j | \mathbf{x}) = 0 \quad i \neq j$
4. $\varepsilon_i | \mathbf{x} \sim N(0, \sigma^2)$

For values $(a, b_1, \dots, b_k)$ of the unknown coefficients $(\alpha, \beta_1, \dots, \beta_k)$ define the Residual Sum of Squares (RSS) as

$$RSS = \sum_{i=1}^n (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki})^2 \quad (2)$$

and we wish to find those values of $a, b_1, b_2, \dots, b_k$ which minimise this expression. Consequently, we differentiate the RSS with respect to each of the parameters, $a, b_1, b_2, \dots, b_k$ and set the resultant derivatives equal to zero.

Differentiating equation (2) with respect to a and setting the expression to zero

$$\begin{aligned} \frac{\partial \left(\sum_{i=1}^n (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki})^2 \right)}{\partial a} &= -2 \sum_{i=1}^n (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki}) \\ &= -2 \sum_{i=1}^n e_i = \sum_{i=1}^n e_i = 0 \end{aligned} \quad (3)$$

And this is the same expression as equation (3) from Handout 1.

Differentiating equation (2) with respect to b_1 and setting the expression to zero

$$\begin{aligned} \frac{\partial \left(\sum_{i=1}^n (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki})^2 \right)}{\partial b_1} &= -2 \sum_{i=1}^n x_{1i} (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki}) \\ &= -2 \sum_{i=1}^n x_{1i} e_i = \sum_{i=1}^n x_{1i} e_i = 0 \end{aligned} \quad (4)$$

And this is similar to the expression in equation (5) in Handout 1.

Differentiating equation (2) with respect to b_2 and setting the expression to zero

$$\begin{aligned} \frac{\partial \left(\sum_{i=1}^n (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki})^2 \right)}{\partial b_2} &= -2 \sum_{i=1}^n x_{2i} (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki}) \\ &= -2 \sum_{i=1}^n x_{2i} e_i = \sum_{i=1}^n x_{2i} e_i = 0 \end{aligned} \quad (5)$$

Finally differentiating equation (2) with respect to b_k and setting the expression to zero

$$\begin{aligned} \frac{\partial \left(\sum_{i=1}^n (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki})^2 \right)}{\partial b_k} &= -2 \sum_{i=1}^n x_{ki} (y_i - a - b_1 x_{1i} - b_2 x_{2i} - \dots - b_k x_{ki}) \\ &= -2 \sum_{i=1}^n x_{ki} e_i = \sum_{i=1}^n x_{ki} e_i = 0 \end{aligned} \quad (6)$$

NOTE

- (i) Equation (3) implies that the residuals sum to zero (providing there is an intercept in the model)
- (ii) Equation (4) implies that the covariance (and hence correlation) between the residuals and the explanatory variable x_1 is zero, that is, they are orthogonal.
- (iii) Equation (5) implies that the covariance (and hence correlation) between the residuals and the explanatory variable x_2 is zero, that is, they are orthogonal.
- (iv) Equation (6) implies that the covariance (and hence correlation) between the residuals and the explanatory variable x_k is zero, that is, they are orthogonal.

The degrees of freedom (DoF) for this model is the number of observation, n , less the number of restrictions imposed on the residuals, which are:

$$\underbrace{\sum_{i=1}^n e_i = 0, \quad \sum_{i=1}^n x_{1i}e_i = 0, \quad \sum_{i=1}^n x_{2i}e_i = 0, \quad \sum_{i=1}^n x_{3i}e_i = 0 \dots, \quad \sum_{i=1}^n x_{ki}e_i = 0}_{k+1}$$

That is, DoF is $(n - k - 1)$.

For the multiple regression model it is not easy to write out an explicit formula for the OLS estimators, a , b_1 , $b_2, \dots, b_k$ (without using matrix algebra) - although solving the $(k + 1)$ equations (3), (4), (5) and (6) simultaneously yields your OLS estimates. One can use the idea of partitioned regression in order to get an algebraic expression for these estimators, suppose, for example, one wants the OLS coefficient estimator on x_1 . Then we can obtain an expression for the OLS estimator of β_1 as:

1. Run the regression $y_i = \delta_0 + \delta_2 x_{2i} + \dots + \delta_k x_{ki} + \varepsilon_{yi}$ by OLS and save the OLS residuals, denoted, $\tilde{y}_i$, where $\tilde{y}_i = y_i - (\hat{\delta}_0 + \hat{\delta}_2 x_{2i} + \dots + \hat{\delta}_k x_{ki})$
2. Run the regression $x_{1i} = \gamma_0 + \gamma_2 x_{2i} + \dots + \gamma_k x_{ki} + \varepsilon_{x1i}$, by OLS and save the OLS residuals denoted, $\tilde{x}_{1i}$, where $\tilde{x}_{1i} = x_{1i} - (\hat{\gamma}_0 + \hat{\gamma}_2 x_{2i} + \dots + \hat{\gamma}_k x_{ki})$
3. Run the regression: $\tilde{y}_i = \alpha + \beta_1 \tilde{x}_{1i} + \varepsilon_i$. As this is a 2-variable regression we know that

$$b_1 = \frac{\sum_{i=1}^n (\tilde{y}_i - \bar{\tilde{y}})(\tilde{x}_{1i} - \bar{\tilde{x}}_1)}{\sum_{i=1}^n (\tilde{x}_{1i} - \bar{\tilde{x}}_1)^2} = \frac{\sum_{i=1}^n \tilde{y}_i \tilde{x}_{1i}}{\sum_{i=1}^n \tilde{x}_{1i}^2}$$

as $\bar{\tilde{y}} = \bar{\tilde{x}}_1 = 0$ and $V(b_1) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{x}_{1i}^2}$ where $\sum_{i=1}^n \tilde{x}_{1i}^2$ is the variation in the variable x_1 , not accounted for by the other variables x_2 through to x_k .

We know that $(1 - R^2)TSS = RSS$.

In which case $(1 - R_{x_1}^2) \sum_{i=1}^n (x_{1i} - \bar{x}_1)^2 = \sum_{i=1}^n \tilde{x}_{1i}^2$, where $R_{x_1}^2$ is the R^2 the regression of x_{1i} on x_2 through to x_k .

Therefore as we know that $R_{x_1}^2 \leq 1$ this implies $\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2 \geq \sum_{i=1}^n \tilde{x}_{1i}^2$, which implies that the variance of the estimator in the multiple regression is greater than that in 2-variable regression $V(b_1) \geq V(b)$.

The estimator for σ^2 (variance of the disturbance term, ε_i) is estimated as:

$$s^2 = \frac{\sum_{i=1}^n e_i^2}{DoF} = \frac{RSS}{DoF} \quad (7)$$

where DoF are the degrees of freedom for the residuals.

In general, the variance-covariance matrix for the OLS coefficient estimators would be:

$$\begin{bmatrix} V(a) & \text{cov}(a, b_1) & \dots & \text{cov}(a, b_k) \\ \text{cov}(a, b_1) & V(b_1) & \dots & \text{cov}(b_1, b_k) \\ \vdots & \vdots & \vdots & \vdots \\ \text{cov}(a, b_k) & \text{cov}(b_1, b_k) & \dots & V(b_k) \end{bmatrix}$$

3 Properties of OLS estimators

These OLS estimators have a number of desirable properties (Gauss Markov Theorem) in that the estimators are BLUE:

- (i) Best - have the minimum variance, such that $V(b_j|\mathbf{x}) \leq V(b_j^*|\mathbf{x})$, $j = 1, \dots, k$, where b_j^* is any alternative unbiased estimator.
- (ii) Linear – linear function of the error term.
- (iii) Unbiased - $E(a|\mathbf{x}) = \alpha$ and $E(b_j|\mathbf{x}) = \beta_j$, $j = 1 \dots k$.
- (iv) Estimators.

We will not prove these conditions, but the proof of unbiasedness is similar to that in Handout 1. Based on this we could show that:

$$b_j|\mathbf{x} \sim N(\beta_j, V(b_j))$$

and

$$\frac{DoF}{\sigma^2} s^2 \sim \chi_{DoF}^2$$

.

4 Interpreting coefficients

In the regression model

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

we can interpret the coefficients by partially differentiating the dependent variable with respect to each explanatory variable

$$\frac{\partial y_i}{\partial x_{1i}} = \beta_1 = \left. \frac{\text{Change in } y_i}{\text{Unit increase in } x_1} \right|_{\text{Holding all else constant}}$$

We can also think of examples where the model is clearly nonlinear in the underlying variables, but is still linear in parameters enabling us to use OLS:

$$1. \quad y_i = \alpha + \beta_1 \ln(x_{1i}) + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i$$

$$\frac{\beta_1}{100} = \left. \frac{\text{Change in } y_i}{1\% \uparrow \text{ in } x_{1i}} \right|_{\text{Holding all else constant}}$$

$$2. \quad \ln(y_i) = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i$$

$$100\beta_1 \approx \left. \frac{\% \text{ change in } y_i}{1 \uparrow \text{ in } x_{1i}} \right|_{\text{Holding all else constant}} \quad (\text{if } \beta_1 \text{ is small})$$

alternatively one can always calculate (and this holds irrespective of the magnitude of β_1)

$$100(\exp(\beta_1) - 1) = \left. \frac{\% \text{ change in } y_i}{1 \uparrow \text{ in } x_{1i}} \right|_{\text{Holding all else constant}}$$

$$3. \quad \ln(y_i) = \alpha + \beta_1 \ln(x_{1i}) + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i$$

$$\beta_1 = \left. \frac{\% \text{ change in } y_i}{1\% \uparrow \text{ in } x_i} \right|_{\text{Holding all else constant}}$$

$$4. \quad y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{1i}^2 + \beta_3 x_{3i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad (\text{the relationship is quadratic } x_{1i})$$

$$\frac{\partial y_i}{\partial x_{1i}} = \beta_1 + 2\beta_2 x_{1i} = \left. \frac{\text{Change in } y_i}{\text{Unit increase in } x_1} \right|_{\text{Holding all else constant}}$$

(response varies linearly in x_{1i}).

$$5. \quad y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{1i} \times z_i + \beta_3 x_{3i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad (\text{there is a cross product term})$$

$x_1 \times z)$

$$\frac{\partial y_i}{\partial x_{1i}} = \beta_1 + \beta_2 z_i = \frac{\text{Change in } y_i}{\text{Unit increase in } x_1} \bigg|_{\text{Holding all else constant}}$$

(response varies linearly in z_i).

5 Explanatory Power of Equation

For the regression model

$$y_i = a + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki} + e_i$$

where $a, b_j, j = 1, \dots, k$ are the OLS parameter estimates and e_i the residuals, then the fitted (Predicted) values for y_i are written as

$$\hat{y}_i = a + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki}$$

Therefore

$$y_i = \hat{y}_i + e_i$$

subtracting $\bar{y}$ from both sides of this equation we have:

$$(y_i - \bar{y}) = (\hat{y}_i - \bar{y}) + e_i$$

Squaring this equation we have

$$(y_i - \bar{y})^2 = (\hat{y}_i - \bar{y})^2 + e_i^2 + 2(\hat{y}_i - \bar{y})e_i$$

Taking sums we write

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n e_i^2 + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y})e_i$$

But the last term is zero as

$$\sum_{i=1}^n (\hat{y}_i - \bar{y})e_i = b_1 \sum_{i=1}^n (x_{1i} - \bar{x}_1)e_i + b_2 \sum_{i=1}^n (x_{2i} - \bar{x}_2)e_i + \dots + b_k \sum_{i=1}^n (x_{ki} - \bar{x}_k)e_i$$

and we know $cov(x_{ji}, e_t) = \sum_t (x_{ji} - \bar{x}_j)e_t = 0$ (from equations 4, 5, 6) Therefore

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n e_i^2 \Rightarrow TSS = ESS + RSS \quad (8)$$

and hence

$$0 \leq R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \leq 1 \quad (9)$$

However, R^2 suffers from the problem that as you add more variables into the equation R^2 will never decrease (as RSS must fall or remain unchanged as more variables

are added into the model) and therefore a policy of maximising R^2 will entail over-parameterising the model.

Consequently, people sometime use other criteria for choosing models:

$$\overline{R}^2 = 1 - \frac{RSS/DoF}{TSS/(n-1)}$$

in this case, while RSS will fall as more variables are added, DOF will also fall as more variables are added: so $\overline{R}^2$ may either increase or decrease. $\overline{R}^2$ will only increase if an F-test on the joint significance on the new variables is in excess of unity (see Appendix A for a proof).

Other criteria people use for choosing between models:

- Akaike Information Criterion: $AIC = 2k/n + \ln(RSS/n)$
- Schwarz (Bayesian) Information Criteria: $BIC = k \ln(n)/n + \ln(RSS/n)$
- Hannan-Quinn Information Criteria: $HQC = 2k \ln(\ln(n))/n + \ln(RSS/n)$

Where the AIC has the weakest punishment for each additional parameters, $(2/n)$, and BIC has the strongest $\ln(n)/n$, provided that $n \geq 8$.

6 Hypothesis testing

6.1 The t-test

We follow a 5-step procedure:

- 1 $H_0 : \beta_j = \beta_{j0}$
- 2 $H_0 : \beta_j \neq \beta_{j0}$
- 3 Choose some appropriate significance level of c , and find the corresponding value from the t-distribution, denoted $-t_{DoF}^{c/2}$ and $t_{DoF}^{c/2}$, where DoF are the degree of freedom from the model, that is, n minus the number of restrictions on the residuals, that is, $n - (k + 1)$.
- 4 $t_j = \frac{b_j - \beta_{j0}}{s_{b_j}} \sim t_{DoF}$, where s_{b_j} = standard error of b_j (in replacing σ^2 by s^2 in the standard error of b_j , we are essentially scaling a $N(0, 1)$ by a $\sqrt{\chi_{DoF}^2 / DoF}$ and this yields a t_{DoF} distribution.
- 5 If t_j is either less than $-t_{DoF}^{c/2}$, or greater than $t_{DoF}^{c/2}$, then we have observed an event which occurs with a probability of less than α and should therefore reject H_0 . The decision rule is: Reject H_0 if $t_j = \left| \frac{b_j - \beta_{j0}}{s_{b_j}} \right| > t_{DoF}^{c/2}$; Do not reject H_0 if $t_j = \left| \frac{b_j - \beta_{j0}}{s_{b_j}} \right| < t_{DoF}^{c/2}$

Consider now the more complex problem where we wish to test the null hypothesis $H_0 : \beta_j - \beta_l + 2\beta_m = D_0$, we again follow the 5-step procedure outlined above, but in this case the actual test statistics is:

$$t = \frac{b_j - b_l + 2b_m - D_0}{\sqrt{V(b_j) + V(b_l) + 4V(b_m) - 2cov(b_j, b_l) + 4cov(b_j, b_m) - 4cov(b_l, b_m)}} \sim t_{DoF}^{c/2}$$

6.2 The F-test

We can also test multiple restrictions:

- 1 $H_0 : \beta_1 = \beta_1^0, \dots, \beta_d = \beta_d^0$
- 2 $H_0 : \beta_j \neq \beta_j^0, j = 1, \dots, d$
- 3 Choose some appropriate significance level of c , and find the corresponding value from the F-distribution, denoted $F_{d, DoF}^c$.

4

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^u/DoF} \sim F_{d,DoF} \quad (10)$$

where $RSS^R = \sum_{i=1} (y_i - a - b_1^0 x_{1i} \dots - b_d^0 x_{di} - b_{d+1} x_{d+1i} \dots - b_k x_{ki})^2$, is the RSS from the restricted model, i.e. the model evaluated at the restricted parameters (under H_0), and $RSS^U = \sum_{i=1} (y_i - a - b_1 x_{1i} \dots - b_d x_{di} \dots - b_k x_{ki})^2$, is the RSS from the model which imposes no restrictions on the parameter estimates, d is the number of restrictions under the null hypothesis, i.e. the number of equality signs in H_0 (and MUST also be equal to the difference in the number of parameters to be estimated in the restricted and unrestricted regression models), DoF is the degrees of freedom associated with the unrestricted regression model (and is n minus the number of parameters estimated in the unrestricted model ($k + 1$)).

5 If F is greater than $F_{d,DoF}^c$, then we have observed an event which occurs with a probability of less than α and should therefore reject H_0 . The decision rule is: Reject H_0 if $F > F_{d,DoF}^c$; Do not reject H_0 if $F < F_{d,DoF}^c$.

NOTE

The F test is always positive as $RSS^R \geq RSS^U$ and this is shown in Appendix B. Providing the dependent variable is the same in the restricted and unrestricted equations then one can always construct the F-test in terms of the R^2 from the restricted and unrestricted models.

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^u/DoF} = \frac{\left(\frac{RSS^R}{TSS} - \frac{RSS^U}{TSS}\right)/d}{\frac{RSS^u}{TSS}/DoF}$$

$$F = \frac{((1 - R_R^2) - (1 - R_U^2))/d}{(1 - R_U^2)/DoF} = \frac{(R_U^2 - R_R^2)/d}{(1 - R_U^2)/DoF} \quad (11)$$

and so equations (10) and (11) must yield exactly the same value for the F-test.

As an example where equation (11) is not the appropriate for undertaking an F-test, consider the unrestricted model:

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \varepsilon_i, \quad i = 1, 2, \dots, n \quad (12)$$

then for testing $H_0 : \beta_2 = 0.5, \beta_3 = 0$, the restricted model is:

$$y_i = \alpha + \beta_1 x_{1i} + 0.5x_{2i} + \beta_4 x_{4i} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

However, the only way of imposing a known (non-zero) coefficient on any variable is to take the variable over to the left hand side as coefficients are only estimated on

variables on the right hand side. Consequently, the restricted equation is:

$$y_i - 0.5x_{2i} = \alpha + \beta_1x_{1i} + \beta_4x_{4i} + \varepsilon_i, \quad i = 1, 2, \dots, n \quad (13)$$

The restricted equation (13) has a different dependent variable to the unrestricted equation (12), in which case, while one can test the null hypothesis on the basis of the F-test in equation (10), an F-test based on equation (11) will not be correct.

6.3 F-test of overall significance

1. $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$
2. $H_1 : \text{Any } \beta_j \neq 0$
3. Choose some appropriate significance level of c , and find the corresponding value from the F-distribution, denoted $F_{k, DoF}^c$.
4. $F = \frac{(RSS^R - RSS^U)/k}{RSS^U/DoF} \sim F_{k, DoF}$, where $RSS^R = \sum_{i=1}^n (y_i - a)^2$, (so $RSS^R = TSS$) and $RSS^U = \sum_{i=1}^n (y_i - a - b_1x_{1i} - \dots - b_kx_{ki})^2$, k is the number of coefficients excluding the intercept and DoF is $n - (k + 1)$. Therefore, in this case, $F = \frac{(TSS - RSS^U)/k}{RSS^U/DoF}$ and dividing top and bottom by TSS we get

$$F = \frac{\left(1 - \frac{RSS^U}{TSS}\right)/k}{\left(\frac{RSS^U}{TSS}\right)/DoF} = \frac{R_U^2/k}{(1 - R_U^2)/DoF}$$

5. If F is greater than $F_{k, DoF}^c$, then we have observed an event which occurs with a probability of less than α and should therefore reject H_0 . The decision rule is: Reject H_0 if $F > F_{k, DoF}^c$; Do not reject H_0 if $F < F_{k, DoF}^c$.

Note:

Appendix C has a labelled Stata output for a multiple regression and Appendix D works through a series of questions in relation to multiple regression analysis.

Appendices

A Condition when $\bar{R}^2$ rises

Consider the two models

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i \quad (\text{A1})$$

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + v_i \quad (\text{A2})$$

where we denote the RSS from each equation as RSS_{A1} and RSS_{A2} , respectively. Then we have that:

$$\bar{R}_{A2}^2 = 1 - \frac{RSS_{A2}/(n-5)}{TSS/(n-1)} \text{ and } \bar{R}_{A1}^2 = 1 - \frac{RSS_{A1}/(n-3)}{TSS/(n-1)}$$

Now if the F statistics to test the null hypothesis $H_0 : \beta_3 = \beta_4 = 0$ is greater than unity, that is, $F \geq 1$. This implies that

$$F = \frac{(RSS_{A1} - RSS_{A2})/2}{RSS_{A2}/(n-5)} \geq 1$$

Rearranging this expression we have

$$(RSS_{A1} - RSS_{A2})/2 \geq RSS_{A2}/(n-5)$$

that is,

$$(n-5)RSS_{A1} \geq (n-5)RSS_{A2} + 2RSS_{A2} \Rightarrow (n-5)RSS_{A1} \geq (n-3)RSS_{A2}$$

$$\Rightarrow RSS_{A1}/(n-3) \geq RSS_{A2}/(n-5)$$

Dividing both sides of the above expression by $TSS/(n-1)$ (which is the same from equation (A1) and equation A2)) as both variables have exactly the same dependent variable, then we have

$$\frac{RSS_{A1}/(n-3)}{TSS/(n-1)} \geq \frac{RSS_{A2}/(n-5)}{TSS/(n-1)}$$

Multiplying both sides of the equation by -1 (and hence swapping the inequality sign around) and adding 1 we have:

$$\bar{R}_{A1}^2 \equiv 1 - \frac{RSS_{A1}/(n-3)}{TSS/(n-1)} \leq 1 - \frac{RSS_{A2}/(n-5)}{TSS/(n-1)} \equiv \bar{R}_{A2}^2$$

That is $\bar{R}^2$ will always fall if you impose a restriction on any model which has an F statistic greater than unity. For a single restriction e.g. $H_0 : \beta_4 = 0$ an $F \geq 1 \Leftrightarrow t \geq 1$.

B Why RSS falls with more variables

Consider the very simple case where we have two models the first of which has no explanatory variables and the second a solitary explanatory variable (the result goes through irrespective of the number of variables (although the algebra does get trickier):

$$y_i = \alpha + \varepsilon_i \quad (\text{B1})$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (\text{B2})$$

Estimating both models by OLS we have:

$$y_i = a + e_i = \hat{y}_i + e_i \quad (\text{B3})$$

$$y_i = (b_0 + b_1 x_i) + e_i = \hat{y}_i + e_i \quad (\text{B4})$$

Corresponding to equations (B1) and (B2), respectively. In either case we can take means of B3 or B4 and we get:

$$\bar{y} = \bar{\hat{y}} + \bar{e} = \bar{\hat{y}}$$

as we estimated the equations by OLS we know that $\bar{e} = 0$. In which case we can always write by subtracting the above equation from (B3) or (B4):

$$y_i - \bar{y} = (\hat{y}_i - \bar{\hat{y}}) + e_i$$

Now if we square both sides of the above equation and take summations we have:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{\hat{y}})^2 + \sum_i e_i^2 + 2 \sum_i (\hat{y}_i - \bar{\hat{y}}) e_i$$

But the last term in the above expression is zero as estimating any equations by OLS we know: $\sum_i x_i e_i = 0$ and as $\hat{y}$ is a function of x (see equation (B4)) this implies $\sum_i \hat{y}_i e_i = 0$, in which case we have:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{\hat{y}})^2 + \sum_i e_i^2 \Rightarrow TSS = ESS + RSS \quad (\text{B5})$$

If we know consider both equation (B1) and equation (B2).

In equation (B1)

$$ESS = \sum_i (\hat{y}_i - \bar{\hat{y}})^2 = \sum_i (a - a)^2 = 0$$

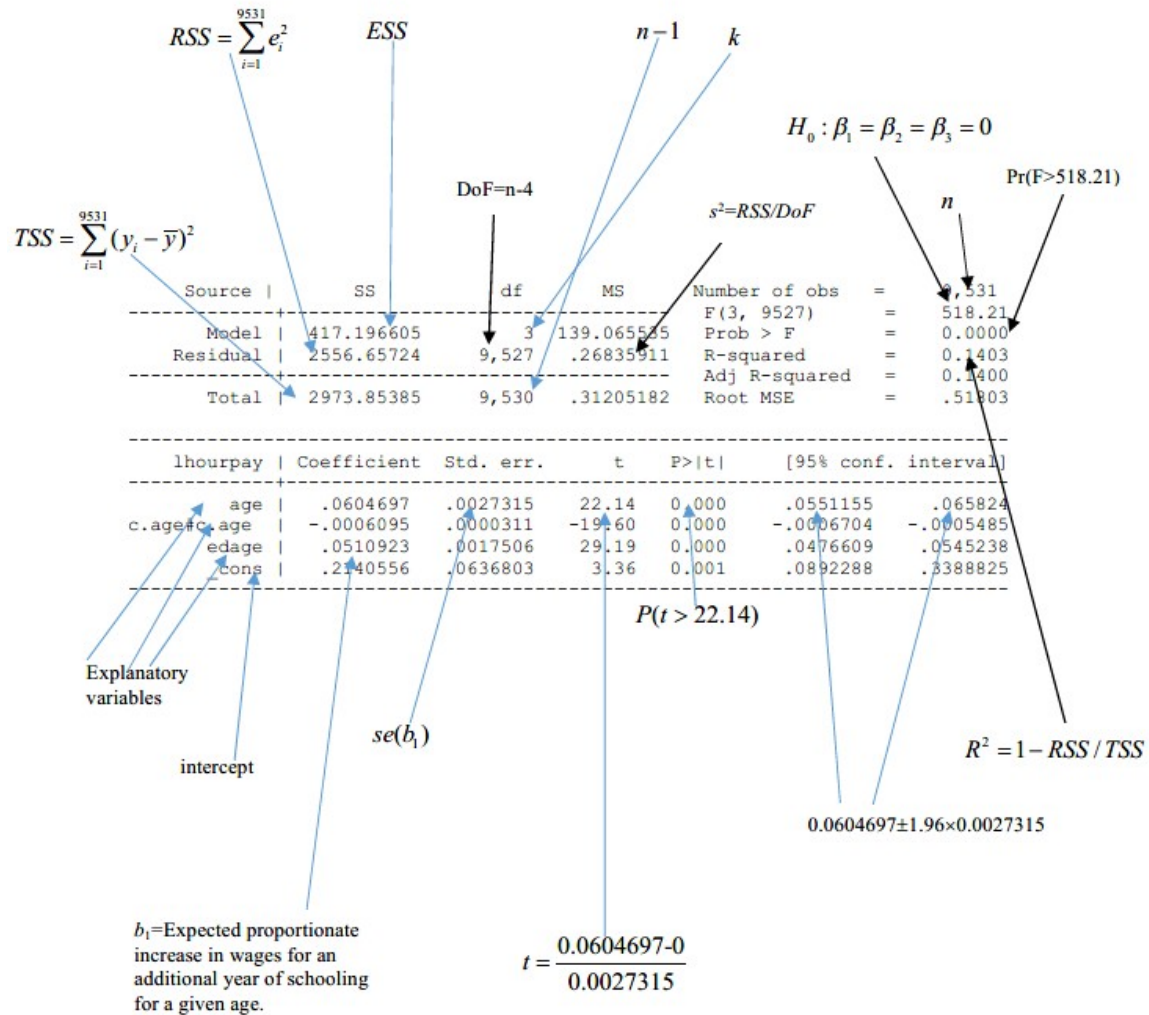
In equation (B2)

$$ESS = \sum_i (\hat{y}_i - \bar{\hat{y}})^2 = \sum_i (a + bx_i - (a + b\bar{x}))^2 = b^2 \sum_i (x_i - \bar{x})^2 > 0$$

In which case $ESS_2 > ESS_1 \Rightarrow RSS_2 < RSS_1$ due to the identity in equation (B5).

C Stata regression output

reg lhourpay age agesq edage if(edage>=0 & edage<=95)



D Exercise questions

Consider the following model

$$\ln(Y_i) = \alpha + \beta_1 x_{1i} + \beta_2 (1/x_{1i}) + \beta_3 x_{2i} + \beta_4 \ln(x_{3i}) + \varepsilon_i$$

(1) The results from an OLS regression estimated using quarterly data over the period 1987:1-2002:4 on equation (1) were

$$\begin{array}{cccccc} \ln(y_i) = 0.034 + 0.164x_{1i} - 0.265(1/x_{1i}) + 0.765x_{2i} + 1.116 \ln(x_{3i}) + e_i \\ (0.054) \quad (0.066) \quad (0.111) \quad (0.337) \quad (0.421) \end{array}$$

$$RSS = 0.1658, R^2 = 0.425$$

- (a) Interpret the coefficients
- (b) At the 5% significance level test the hypotheses (i) $\beta_1 = 0$ and (ii) $\beta_4 = 1$
- (c) Test the hypothesis that the response of $\ln(y_i)$ to a change in x_{1i} is 1, when $x_{1i} = 0.5$, given that $cov(b_1, b_2) = 0.02$.
- (d) Test the hypothesis that all slope coefficients are equal to zero.
- (e) Write out the form of the restricted model, as would be estimated by a computer programme, when $\beta_1 = \beta_4 + 1$ and $\beta_3 = -3\beta_2$. The RSS from this model is 0.1889, test the validity of the restrictions.
- (f) Calculate the RSS for the model when x_{2i} is dropped from the above model.

‘ **Answers**

- (a) $\frac{\partial \ln(Y_i)}{\partial x_{1i}} = \beta_1 - \frac{\beta_2}{x_{1i}^2}$ = Proportionate change in y for a unit increase in x_1 .
 $\frac{\partial \ln(Y_i)}{\partial x_{2i}} = \beta_3$ = Proportionate change in y for a unit increase in x_2 .
 $\frac{\partial \ln(Y_i)}{\partial \ln(x_{3i})} = \beta_4$ = % change in y for a 1% increase in x_2 .

- (b) $H_0 : \beta_1 = 0; H_1 : \beta_1 \neq 0$ $t_{64-5}^{0.025} = \pm 2.001$

$$t = \frac{0.164 - 0}{0.066} = 2.485 \Rightarrow \text{Reject } H_0$$

$$H_0 : \beta_4 = 1; H_1 : \beta_4 \neq 1$$
 $t_{64-5}^{0.025} = \pm 2.001$

$$t = \frac{1.116 - 1}{0.421} = 0.276 \Rightarrow \text{Do not reject } H_0$$

- (c) $H_0 : \beta_1 - \frac{\beta_2}{x_{1i}^2} = 1; H_1 : \beta_1 - \frac{\beta_2}{x_{1i}^2} \neq 1$ $t_{64-5}^{0.025} = \pm 2.001$

$$\text{At } x_{1i} = 0.5 \Rightarrow H_0 : \beta_1 - 4\beta_2 = 1$$

$$t = \frac{0.164 - 4(-0.265) - 1}{\sqrt{0.066^2 + 16(0.111)^2 - 8(0.02)}} = \frac{1.224 - 1}{\sqrt{0.041492}} = 1.100 \Rightarrow \text{Do not reject } H_0$$

- (d) $H_0 : \beta_1 = \beta_2 = \dots \beta_4 = 0; H_0 : \beta_j \neq 0 \forall j = 1, 2, 3, 4$ $F_{4,59}^{0.05} \approx 2.53$

$$F = \frac{0.425/4}{(1 - 0.425)/59} = 10.90 \Rightarrow \text{Reject } H_0$$

- (e)

$$\ln(y_i) = \alpha + (\beta_4 + 1)x_{1i} + \beta_2(1/x_{1i}) - 3\beta_2x_{2i} + \beta_4 \ln(x_{3i}) + \varepsilon_i$$

$$\ln(y_i) - x_{1i} = \alpha + \beta_2[(1/x_{1i}) - 3x_{2i}] + \beta_4(\ln(x_{3i}) + x_{1i}) + \varepsilon_i$$

$$H_0 : \beta_1 = \beta_4 + 1, \beta_3 = -3\beta_2; H_1 : \beta_1 \neq \beta_4 + 1, \beta_3 \neq -3\beta_2, F_{2,59}^{0.05} = 3.153$$

$$F = \frac{(0.1889 - 0.1658)/2}{0.1658/59} = 4.110 \Rightarrow \text{Reject } H_0$$

Note in this case we cannot use the formula

$$F = \frac{(R_U^2 - R_R^2)/d}{(1 - R_U^2)/DoF}$$

to calculate the F test as the dependent variable is different between the unrestricted model, $\ln(Y_i)$, and the restricted model, $\ln(Y_i) - x_{1i}$.

(f) $H_0 : \beta_3 = 0$; $H_1 : \beta_3 \neq 0$, $t_{64-5}^{0.025} = \pm 2.001$

$$t = \frac{0.765 - 0}{0.337} = 2.27 \Rightarrow F = 5.153 = \frac{(RSS^R - 0.1658)/1}{0.1658/59} \Rightarrow RSS^R = 0.1803$$

E Testing linear restrictions

When testing for linear restrictions in an OLS model the test can be obtained from comparing the RSS from the restricted and unrestricted model (see equation (10)). However, Stata constructs the test using the vector of unrestricted coefficient estimates and the corresponding variance-covariance matrix.

Imagine you have a model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i$$

in matrix form we can write this model as:

$$y = X\beta + \varepsilon$$

$$\text{where } y = \begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{pmatrix}, x = \begin{pmatrix} 1 & x_{11} & x_{21} & x_{31} \\ 1 & x_{12} & x_{22} & x_{32} \\ \dots & \dots & \dots & \dots \\ 1 & x_{1n} & x_{2n} & x_{3n} \end{pmatrix}, \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \text{ and } \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{pmatrix}$$

Suppose you want to test $H_0 : \beta_1 = 0; \beta_2 = 0$ then define two new matrices:

$$R = \begin{pmatrix} 0, 1, 0, 0 \\ 0, 0, 1, 0 \end{pmatrix} \text{ and } r = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\text{Now consider the expression } (R\beta - r) = \begin{pmatrix} 0, 1, 0, 0 \\ 0, 0, 1, 0 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \beta_1 - 0 \\ \beta_2 - 0 \end{pmatrix}$$

which is what we were testing in the null hypothesis. The F-test is then written as:

$$F = (R \times b - r)' [R \times V(b) \times R']^{-1} (R \times b - r) / 2$$

where b is the vector of OLS coefficient estimates and $V(b)$ is the OLS variance-covariance matrix.

Consider the elements of the F-test above we have:

$$(Rb - r) = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$R \times V(b) \times R' = \begin{pmatrix} 0, 1, 0, 0 \\ 0, 0, 1, 0 \end{pmatrix} \begin{pmatrix} V(b_0) & cov(b_0, b_1) & cov(b_0, b_2) & cov(b_0, b_3) \\ cov(b_0, b_1) & V(b_1) & cov(b_1, b_2) & cov(b_1, b_3) \\ cov(b_0, b_2) & cov(b_1, b_2) & V(b_2) & cov(b_2, b_3) \\ cov(b_0, b_3) & cov(b_1, b_3) & cov(b_2, b_3) & V(b_3) \end{pmatrix} \begin{pmatrix} 0, 0 \\ 1, 0 \\ 0, 1 \\ 0, 0 \end{pmatrix}$$

$$R \times V(b) \times R' = \begin{pmatrix} V(b_1) & cov(b_1, b_2) \\ cov(b_1, b_2) & V(b_2) \end{pmatrix}$$

Putting these expressions together the F-test is then written as:

$$F = \begin{pmatrix} b_1 & b_2 \end{pmatrix} \begin{pmatrix} V(b_1) & cov(b_1, b_2) \\ cov(b_1, b_2) & V(b_2) \end{pmatrix}^{-1} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} / 2$$

In the special case in which $cov(b_1, b_2) = 0$ we have:

$$F = \frac{\left[\frac{b_1^2}{V(b_1)} + \frac{b_2^2}{V(b_2)} \right]}{2} = \frac{(t_1^2 + t_2^2)}{2}$$

where t_j is the t-test for $H_0 : \beta_j = 0$, as soon as $cov(b_1, b_2) \neq 0$ then you will get a different result from doing an F-test as opposed to a series of individual t-tests.

F Stata regression commands

To run a regression of lhourpay against edage and a quadratic in age and then do some hypothesis testing on this model, we type:

```
regress lhourpay age c.age#c.age edage if(edage>=0 & edage<=95)
scalar rss=e(rss)
scalar dof=e(df_r)
/* Testing all slope coefficients are zero */
test age c.age#c.age edage
/* Testing the two coefficients on age are zero */
test age c.age#c.age
/* Testing the coefficient on edage is 0.1 */
test edage=0.1
/* Testing the coefficient on age -100 times that on age^2 */
test age + 100*c.age#c.age=0
/* Testing the coefficient on age -100 times that on age^2 and the
coefficient on edage is zero*/
test (age + 100*c.age#c.age=0) (edage=0)
/* Some of these tests could have been done by hand e.g. testing
the coefficient on edage=0.1 or the coefficeint on age is -100 that on
age^2*/
matrix b2=e(b)
matrix vb2=e(V)
scalar ttest1=(b2[1,3]-0.1)/sqrt(vb2[3,3])
scalar ttest2=(b2[1,1]+100*b2[1,2])/sqrt(vb2[1,1]+100^2*vb2[2,2]+2*100*vb2[1,2])
scalar list ttest1 ttest2
/* Testing the coeffcients on age and age^2 are zero */
regress lhourpay edage if(edage>=0 & edage<=95)
scalar rss2=e(rss)
scalar dof2=e(df_r)
scalar ftest1=(rss2-rss1)/(dof2-dof1)/(rss1/dof1)
scalar list ftest1
/* Test matrix equaivalent to this test is written as (see */
matrix R=(1, 0, 0, 0 \ 0, 1, 0, 0)
matrix r=(0\ 0)
matrix ftest11=(R*b2'-r)'\*inv(R*vb2*R')*(R*b2'-r)/2
matrix list ftest11
```

G R regression commands

To run a regression of lhourpay against edage and a quadratic in age and then do some hypothesis testing on this model, we type:

```
#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:
```

```
library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)
```

```
model3 <- lm(lhourpay ~ age + I(age^2) + edage, data=subset(qlfs18,
  edage1 >= 0 & edage1 <= 95))
summary(model3)
res3 <- resid(model3)
yhat3 <- fitted(model3)
rss3 <- sum(res3^2)
n3 <- nobs(model3)
rsquared3 <- summary(model3)$r.squared
k3 <- length(model3$coefficients)-1
df3 <- n3 - k3 - 1
b3 <- coef(model3)
```

```

vb3 <- vcov(model3)
# Testing all slope coefficients are zero
test3_1 <- linearHypothesis(model3, c("age = 0", "I(age^2) = 0", "edage = 0"),
                             test="F")
fstat3_1 <- test3_1$F[2]
# Testing the two coefficients on age and age^2 are zero using RSS formula*/
test3_2 <- linearHypothesis(model3, c("age = 0", "I(age^2) = 0"), test="F")
fstat3_2 <- test3_2$F[2]
# Testing the two coefficients on age and age^2 are zero using variance-covariance
# matrix formula*/
test3_2b <- linearHypothesis(model3, c("age = 0", "I(age^2) = 0"), test="F",
                             vcov = vb3)
fstat3_2b <- test3_2b$F[2]
# Testing the coefficient on edage is 0.1 */
test3_3 <- linearHypothesis(model3, c("edage = 0.1"), test="F")
fstat3_3 <- test3_3$F[2]
# Testing the coefficient on age is -100 times that on age^2 */
test3_4 <- linearHypothesis(model3, c("age + 100 * I(age^2) = 0"), test="F")
fstat3_4 <- test3_4$F[2]
# Testing the coefficient on age -100 times that on age^2 and the coefficient
# on edage is zero*/
test3_5 <- linearHypothesis(model3, c("age + 100 * I(age^2) = 0", "edage=0"),
                             test="F")
fstat3_5 <- test3_5$F[2]
# Some of these tests could have been done by hand e.g. testing the coefficient
# on edage=0.1 or the coefficient on age is -100 that on age^2*/
fstat3_3b <- ((b3[4]-0.1)/sqrt(vb3[4,4]))^2
fstat3_4b <- ((b3[2]+100*b3[3])/sqrt(vb3[2,2]+100^2*vb3[3,3]+2*100*vb3[2,3]))^2
c(fstat3_3, fstat3_3b)
c(fstat3_4, fstat3_4b)
# Testing the coefficients on age and age^2 are zero */
model4 <- lm(lhourpay ~ edage, data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model4)
res4 <- resid(model4)
rss4 <- sum(res4^2)
n4 <- nobs(model4)
k4 <- length(model2$coefficients)-1

```

```
df4 <- n4 - k4 - 1
fstat3_2c <- (rss4 - rss3)/(df4-df3)/(rss3/df3)
c(fstat3_2, fstat3_2b, fstat3_2c)
# Test matrix equivalent to this test is written as
R3 <- matrix(c(0, 1, 0, 0, 0, 0, 1, 0), nrow=2, ncol=4, byrow = T)
r3 <- matrix(c(0, 0), nrow=2, ncol=1, byrow = T)
fstat3_2d=t((R3*%b3-r3))%*%solve(R3*%vb3*%t(R3))%*%(R3*%b3-r3)/2
c(fstat3_2, fstat3_2b, fstat3_2c, fstat3_2d)
```

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